

Randomized and low-rank approximation

Optimal pivot count for RPCholesky trace approximation

Difficulty: challenging **Importance:** interesting to the community

Status: Partially resolved

Rating rationale: Challenging because a uniform near-optimal pivot count must exploit adaptive residual structure; community impact is efficient kernel and PSD matrix approximation.

Topic: randomized low-rank approximation; kernel matrices

Further partial result - 29 September 2026

Author: Diar Heidary. *Iterated-logarithmic pivot bounds for randomly pivoted Cholesky*, submitted manuscript, 29 September 2026.

Theorems 1.1 and 1.2 (pp. 3-4) give sufficient pivot counts for the original coordinate, exact-arithmetic RPCholesky law on every finite complex Hermitian positive-semidefinite input, without a spectral-decay or eigenvector assumption:

$$\mathbb{E} \operatorname{tr}(R_k) \leq (1 + \varepsilon) \tau_r(A), \quad k = \min\{n, K_j(r, \varepsilon)\}, \quad j \in \{2, 4\},$$

where, for absolute constants C_2, C_4 and $0 < \varepsilon \leq 1$,

$$K_2(r, \varepsilon) \leq C_2 r(\varepsilon^{-1} + \Lambda_2(r)), \quad K_4(r, \varepsilon) \leq C_4 r(\varepsilon^{-1} + \Lambda_4(r)).$$

Here $\Lambda_1(r) = \log(e + r)$ and $\Lambda_{j+1}(r) = \log(e + \Lambda_j(r))$ regularize the iterated logarithms at small ranks. Explicit finite-rank counts appear in equations (18) and (44); rank one has $K_j(1, \varepsilon) = \lceil \varepsilon^{-1} \rceil$, and zero rank- r tails terminate in at most r positive pivots. Thus the reported unrestricted counts are $O(r/\varepsilon + r \log \log r)$ and, more sharply, $O(r/\varepsilon + r \log \log \log \log r)$ in large-rank notation.

Proposition 8.1 (pp. 19-21) also gives explicit robustness bounds for two-sided multiplicative approximations to the pivot probabilities, with exact Schur updates. **The unrestricted $O(r/\varepsilon)$ target remains open:** the fourfold iterated-logarithmic rank term is still unbounded. The prior tail-envelope result below addresses a different, restricted spectral scope.

The manuscript discloses assistance from OpenAI Codex/ChatGPT and Claude in the mathematical development, exposition, experiments and proof review. This submission records a manuscript result; it does not assert formal verification or external human peer review.

Partial resolution - 13 September 2026

Author: Sidney Holden, Center for Computational Biology, Flatiron Institute, Simons Foundation. [Submission and verified affiliation](#).

[Theorems 4.1, 5.1 and 6.1](#) prove the displayed pivot-count guarantee for the original exact-arithmetic RPCholesky law under these additional tail assumptions. With $a = \lambda_{r+1}(A) > 0$, for every tail index $1 \leq j \leq n - r$:

- $\lambda_{r+j}(A) \leq a2^{-(j-1)}$ gives $C = 3$.
- $\lambda_{r+j}(A) \leq a/j^2$ gives $C = 4$.
- $\lambda_{r+j}(A) \leq a/j^p$ for a fixed $p > 1$ gives an explicit finite C_p depending only on p .

Leading eigenvalues and complex eigenvectors are unrestricted; zero tails and the dimension cap are handled by exact termination. These are tail-normalized hypotheses after rank r , not automatic consequences of global spectral decay. **The unrestricted RA-01 target remains open:** no universal constant for all PSD inputs, or counterexample to all such constants, is established. Theorem 7.1 rules out an auxiliary continuous-time count bound with unit tail-rate coefficient; Proposition 8.1 shows a limitation of the determinant upper comparison. Neither is a counterexample to RA-01.

The stated partial results passed a separate [independent Codex AI-agent informal mathematical audit](#). [Proof source](#). Exact finite checks and optional floating diagnostics passed; these do not replace the general proof. AI assistance in review and submission preparation is disclosed. No Lean verification, external human peer review or historical novelty claim is asserted. The original statement and prior-source credit are retained below.

Context and notation

For a Hermitian positive-semidefinite $A \in \mathbb{C}^{n \times n}$, define the exact-arithmetic RPCholesky residuals by $R_0 = A$. Conditional on $R_t \neq 0$, select j with probability $(R_t)_{jj} / \text{tr}(R_t)$ and set

$$R_{t+1} = R_t - \frac{R_t(:,j)R_t(j,:)}{(R_t)_{jj}}.$$

If $R_t = 0$, keep all subsequent residuals zero. Eigenvalues are ordered $\lambda_1(A) \geq \dots \geq \lambda_n(A) \geq 0$, and $\tau_r(A) = \sum_{j>r} \lambda_j(A)$. This is the pivot rule in Chen, Epperly, Tropp, and Webber, *Randomly pivoted Cholesky: Practical approximation of a kernel matrix with few entry evaluations*, Algorithm 1; their Lemma 5.5 gives the current comparison $\mathbb{E} \text{tr}(R_r) \leq 2^r \tau_r(A)$.

Problem statement

Does there exist a universal constant $C \geq 1$ such that, for every $n \geq 1$, every Hermitian positive-semidefinite $A \in \mathbb{C}^{n \times n}$, every integer $1 \leq r \leq n$, and every $0 < \varepsilon < 1$, RPCholesky satisfies

$$\mathbb{E} \text{tr}(R_k) \leq (1 + \varepsilon) \tau_r(A), \quad k = \min\{n, \lceil Cr/\varepsilon \rceil\}?$$

The expectation is over the algorithm's adaptive pivots. The constant must be independent of dimension, spectrum, rank, and tolerance. The target concerns this fixed pivoting rule.

References

Epperly, *A new analysis of the randomly pivoted Cholesky algorithm*, §1.2, conjecture immediately after Eq. (1.5), and Corollary 1.3. Earlier motivation appears in Epperly, *Make the Most of What You Have*, Caltech dissertation (2025), §11.1, p. 181.

Status check

The August 2026 paper proves the larger bound $k \geq r/\varepsilon + 2r\sqrt{\log r} + r \log(1/\varepsilon) + 2.3r$ and explicitly conjectures the displayed improvement. Searches for the paper's title, *RPCholesky optimal r epsilon*, and *RPCholesky conjecture* found no subsequent resolution. The arXiv record listed only v1, posted August 21, 2026. The weaker dissertation Conjecture 11.1 is therefore excluded as resolved by this later preprint.

Audit — 2026-09-10

Rechecked the [August 2026 conjecture after \(1.5\)](#) and its current arXiv record, still v1. Optimal-pivot and RPCholesky follow-up searches found no resolution. The proved extra rank-dependent term does not establish the uniform $O(r/\varepsilon)$ target.